THE TRANSFORMATION COMPARING THE TWO WAY BASE CHANGE OF THE RELATIVE DUALIZING FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the base change comparison for the derived pushforward and the passage to the mate under adjunction.

Editorial note. The label lemma-flat-precompose-pus names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Cross-chapter same-name check: lemma-flat-precompose-pus is not carded in any other chapter. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-duality.tex –label lemma-flat-precompose-pus –start 801 –end 815 –local-id 28176 returned RESULT=OK.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Base change for the derived pushforward under Tor independence. Derived Categories of Spaces, the lemma comparing the pullback of the derived pushforward of a quasi-coherent complex with the derived pushforward of its pullback, under the Tor independence assumption. Shows that the base change map between the two composites of pullback and pushforward is an isomorphism.
- Uniqueness of the transformation between adjoint functors. The standard observation that a natural transformation between right adjoints is an isomorphism as soon as its mate between the left adjoints is an isomorphism. Transfers the isomorphism of the base change map to the transformation between the relative dualizing functors.

Source reference. Original equation 07HF: sources/stacks-spaces-duality/pinned-original-source.tex, source lines 773–773. Prerequisite pointer invoked at proof line(s) 803; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08IR: sources/stacks-chow/tags.txt, source lines 10824–10824. Prerequisite pointer invoked at proof line(s) 811; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In diagram (07HF) the map $a \circ Rg_* \leftarrow Rg'_* \circ a'$ is an isomorphism.*

Proof. The base change map $Lg^* \circ Rf_*K \to Rf'_* \circ L(g')^*K$ is an isomorphism for every K in $D_{QCoh}(\mathcal{O}_X)$ by Derived Categories of Spaces, Lemma 08IR (this uses the assumption of Tor independence). Thus the corresponding transformation between adjoint functors is an isomorphism as well. □